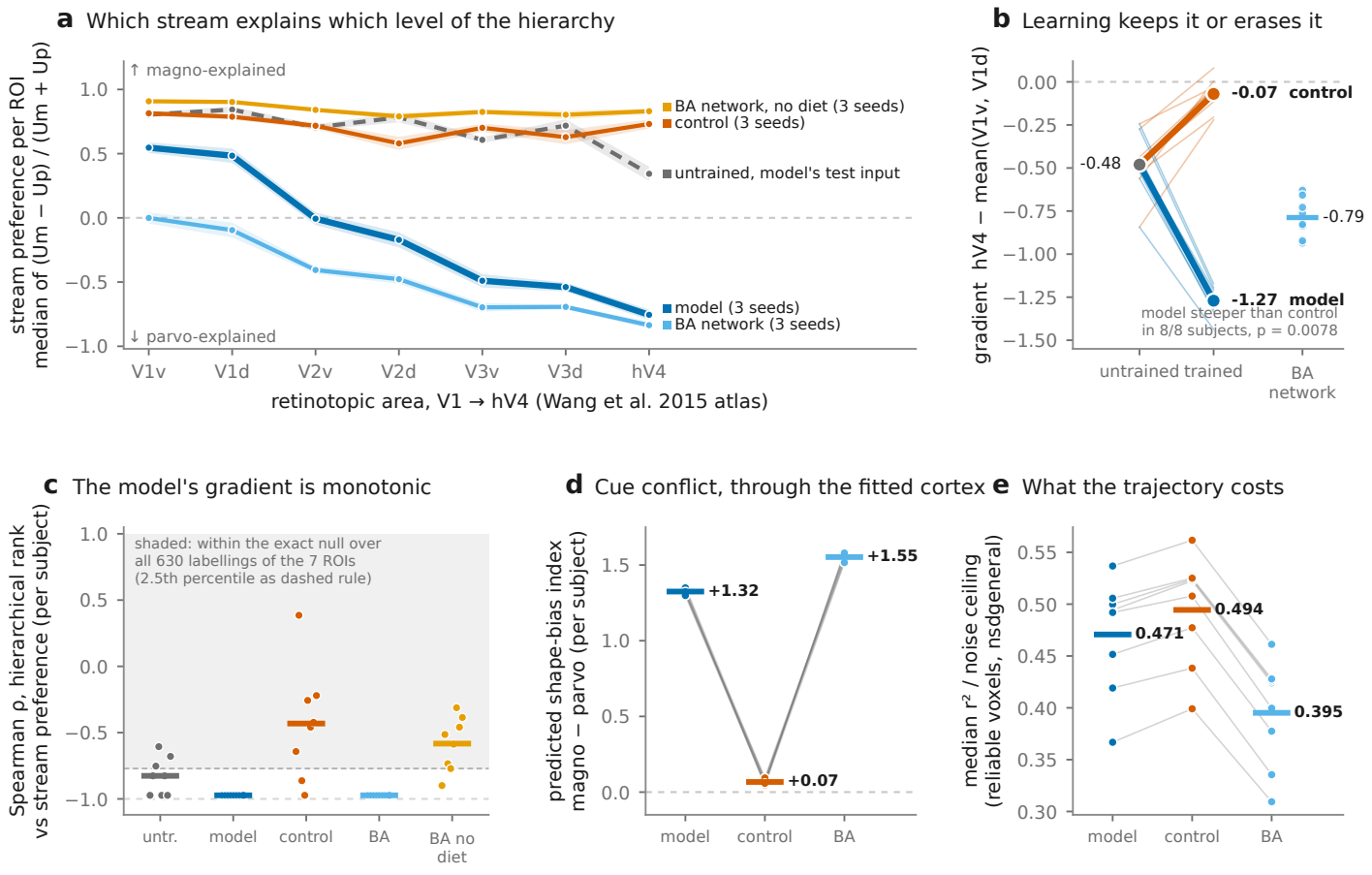
a Which stream explains which level of the hierarchy
↑ magno-explained
↓ parvo-explained
stream preference per ROI
median of (Um − Up) / (Um + Up)
1.0
0.5
0.0
−0.5
−1.0
V1v
V1d
V2v
V2d
V3v
V3d
hV4
retinotopic area, V1 → hV4 (Wang et al. 2015 atlas)
BA network, no diet (3 seeds)
control (3 seeds)
untrained, model's test input
model (3 seeds)
BA network (3 seeds)
b Learning keeps it or erases it
gradient hV4 − mean(V1v, V1d)
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
-0.07 control
-0.48
-0.79
-1.27 model
model steerper than control
in 8/8 subjects, p = 0.0078
untrained trained
BA
network
c The model's gradient is monotonic
Spearman ρ, hierarchical rank
vs stream preference (per subject)
1.0
0.5
0.0
−0.5
−1.0
shaded: within the exact null over
all 630 labellings of the 7 ROIs
(2.5th percentile as dashed rule)
untr.
model
control
BA
BA no
diet
d Cue conflict, through the fitted cortex
predicted shape-bias index
magno − parvo (per subject)
1.5
1.0
0.5
0.0
+1.32
+1.55
+0.07
model
control
BA
e What the trajectory costs
median r² / noise ceiling
(reliable voxels, nsdgeneral)
0.55
0.50
0.45
0.40
0.35
0.30
0.494
0.471
0.395
model
control
BA